

Math 142, S12

Quiz 2

Section:

Name: Answer

SSN: xxx-xx-

Instructions: There is totally 1 problem. You must show all of your work to receive a credit for a correct answer. No calculator is allowed in the quiz and exam.

Problem 1. (10 points) Evaluate the following integrals:

(a) $\int \ln(2x) dx$

Let $u = \ln(2x)$, $dv = dx$,

then $du = \frac{1}{x} dx$, $v = x$

Thus

$$\int \ln(2x) dx = x \ln(2x) - \int x \cdot \frac{1}{x} dx$$

$$= x \ln(2x) - \int dx$$

$$= x \ln(2x) - x + C$$

(b) $\int_0^1 t e^{-t} dt$

Let $f(t) = t$, $g'(t) = e^{-t}$

then $f'(t) = 1$, $g(t) = -e^{-t}$

Thus

$$\int_0^1 t e^{-t} dt = \left[-t e^{-t} \right]_0^1 - \int_0^1 (-e^{-t}) dt$$

$$= [-e^{-1} - 0] + \int_0^1 e^{-t} dt$$

$$= -e^{-1} + (-e^{-t}) \Big|_0^1$$

$$= -e^{-1} + [-e^{-1} + 1]$$

$$= 1 - 2e^{-1} = 1 - \frac{2}{e}$$

(c) $\int \sin^2 x \cos^3 x dx$

$$\int \sin^2 x \cos^3 x dx = \int \sin^2 x \cos^2 x \cos x dx$$

$$= \int \sin^2 x (1 - \sin^2 x) \cos x dx$$

$$\stackrel{u = \sin x}{=} \int u^2 (1 - u^2) du$$

$$= \int (u^2 - u^4) du$$

$$= \frac{u^3}{3} - \frac{u^5}{5} + C$$

$$= \frac{\sin^3 x}{3} - \frac{\sin^5 x}{5} + C$$